

DC MOTOR

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1. DEFINITION

A DC motor is an electrical machine that converts direct current electrical energy into mechanical energy (rotational).

3. WORKING PRINCIPLE

- When a current-carrying conductor is placed in a magnetic field, it experiences a force.
- In a DC motor, the armature conductors carry current. The interaction between the field magnetic field and armature current produces a torque, causing rotation.
- The commutator reverses the current in the armature conductors every half turn to keep the torque unidirectional.

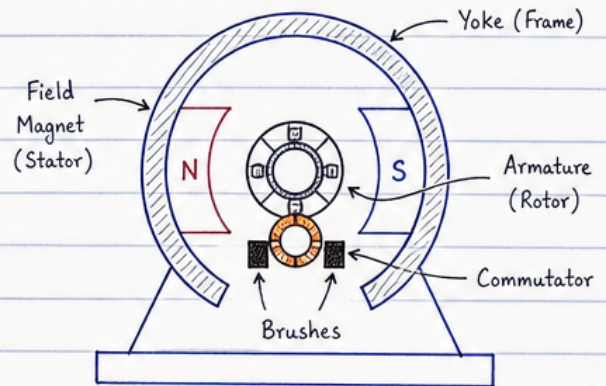
4. IMPORTANT FORMULAS

- ① Back EMF (E_b): $E_b = \frac{P \Phi Z N}{60 A}$ volts
- ② Torque (T): $T = \frac{P \Phi Z I_a}{2 \pi A}$ N.m
- ③ Speed (N): $N = \frac{E_b 60 A}{P \Phi Z}$ rpm
- ④ Motor Equation: $V = E_b + I_a R_a$

5. TYPES OF DC MOTORS

- Separately Excited DC Motor
- Self Excited DC Motor
 - Shunt Motor
 - Series Motor
 - Compound Motor (Cumulative / Differential)

2. CONSTRUCTION (Basic)



P = Number of poles
Φ = Flux per pole (Wb)
Z = Total armature conductors
N = Speed (rpm)
A = Number of parallel paths
I_a = Armature current (A)
R_a = Armature resistance (Ω)
V = Supply voltage (V)
E_b = Back EMF (V)

6. CHARACTERISTICS (General)

- Good starting torque
- Wide speed control
- High efficiency
- Easy to reverse

NOTE: The direction of rotation can be reversed by reversing either the armature connections or the field connections (not both).